

Week 3 Lab Report — Control Flow & Functions

Introduction to Big Data Analytics · AUCA

Overview

The dataset `week3_students.csv` contains 40 student records, of which 3 had broken (non-numeric) values in the score column. My pipeline used a try/except block around `int(parts[7])` for every row: when the conversion succeeded, the score was added to a running total and the valid-record counter increased; when it failed (raising a `ValueError`), the student's ID was logged and a separate bad-record counter increased, without stopping the program. This allowed the pipeline to process all 40 rows safely and produce a clean result from the 37 valid records.

The 3 Broken Records

Student ID	What was wrong
AUCA008	Score field was the text "N/A" instead of a number
AUCA020	Score field was empty/blank
AUCA032	Score field was the text "absent" instead of a number

Results

- Valid records: 37 · Bad records: 3
- Average score (valid only): 75.3
- Passed (≥ 50): 35 · Failed: 2
- Top student: Bonaventure Nkurunziza, score 95, grade A
- District with highest average score: Muhanga (87.0), among 8 districts

Reflection

1. At what percentage would you stop trusting the dataset?

7.5% (3 of 40) feels tolerable to skip and note — it is plausible as ordinary data-entry error. If it climbed toward 15–20% or higher, I would stop treating it as a few typos and start suspecting something systemic, such as a bad export or a broken data-entry form. At that point, skipping rows silently would meaningfully distort the average, so I would go back to the source and investigate rather than continue dropping records.

2. One thing that surprised or confused you:

[Write 2–3 sentences here about which concept from the session took a few tries to click for you — for example the try/except pattern, the dictionary `.get()` default trick, or the ladder-order bug from Question 1.3.]

Closing note: This lab reinforced how try/except turns a program from crashing on bad input into one that keeps going and reports what it could not handle — a small structural change that matters a great deal when working with real-world, messy data.